

9.5.21.

In Morse code, symbols are represented by variable-length sequences of dots and dashes. (For example, A = $\cdot-$, 1 = $\cdot- - -$, ? = $\cdot- - \cdot$.) How many different symbols can be represented by sequences of seven or fewer dots and dashes?

Each position in a Morse-code sequence has exactly 2 choices:

dot ($\cdot$) or dash ($-$)

So for a sequence of exactly n positions, there are 2^n different sequences, and we must show how many different symbols can be represented by sequences of seven or fewer dots and dashes. Therefore:

$$\begin{aligned} 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6 + 2^7 = \\ 2 + 4 + 8 + 16 + 32 + 64 + 128 = \boxed{254} \end{aligned}$$

Solution: There're 254 different symbols represented by sequences of seven or fewer dots and dashes.